Product Design ,Processes And Functions

Product & Service Design

- The process of deciding on the unique characteristics of a company’s product & service offerings
- Product Design deals with form and function of a product.
- Form Design — is associated with product's shape.
- Functional design — is associated with product's working

Products versus Services

- Products:
  - Tangible offerings
  - Dimensions, materials, tolerances & performance standards
- Services:
  - Intangible offerings
  - Physical elements + sensory, esthetic, & psychological benefits

Principles of good product design

- From customer prospective
- It should function correctly
- It should have required standard of reliability
- It should be easy to operate
- It should have easy accessibility for servicing
- It should obtain good space utilization
- It Should have pleasant appearance
- It Should be reasonably priced From Manufacturer prospective (for making adequate profit)
- It should be Easy to manufacture using standard components
- Well designed with minimum number of parts Minimum number of operations
- Easy to pack and distribute

Principles of good product design 2

Factors affecting Product Design

1. Technical Factors
2. Industrial Design Factors
3. Economic Factors

Technical Factors

Operating conditions

Skill of workers

Environmental Factors

Length of Time of Production

Type of Materials Used

Industrial Factors

Function

Will product function at minimum cost?

Ergonomics

Is the product suitable for human use?

Does the product cause excessive fatigue to the workers?

Will product function at minimum cost?

Appearance

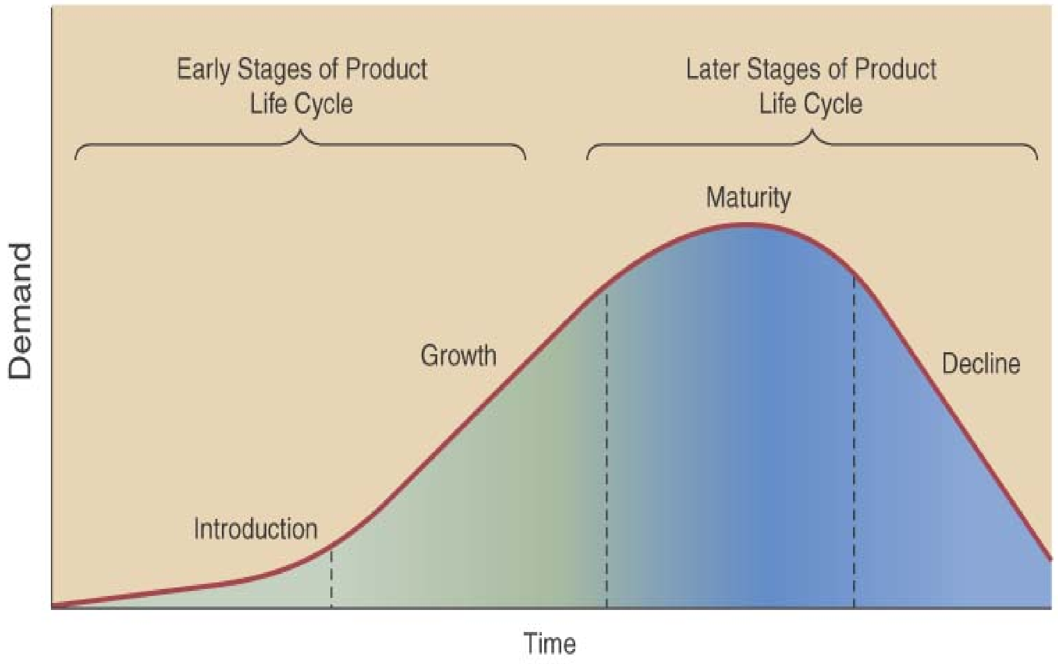
Does the product have pleasing appearance? Does it create esteem?

Economic Factors...

1. Materials, Material specification, Yield ,Quantity
2. Methods-Equipment ,Layout, Labor, Tolerance Tooling
3. Standards- Is design simple? Are types and verities minimum? Does it use standard parts?
4. Finish -Is the right finish being used consistent with cost, endurance and appearance requirements?

Product Life Cycle

- **Product life cycle stages**
  - **Introduction**
  - **Growth**
  - **Maturity**
  - **Decline**
- **Facility & process investment depends on life cycle**



Extending the Product Life Cycle

- Extension strategies extend the life of the product before it goes into decline.
- Examples of the techniques are: Advertising — try to gain a new audience or remind the current audience
- Price reduction — more attractive to customers
- Adding value — add new features to the current product, e.g. improving the specifications on a smartphone
- Explore new markets — selling the product into new geographical areas or creating a version targeted at different segments
- New packaging — brightening up old packaging or subtle changes

Steps in Product Design

- Idea Development:
  - A need is identified & a product idea to satisfy it is put together
- Product Screening:
  - Initial ideas are evaluated for difficulty & likelihood of success
- Preliminary Design & Testing
  - Market testing & prototype development
- Final Design
  - Product & service characteristics are set

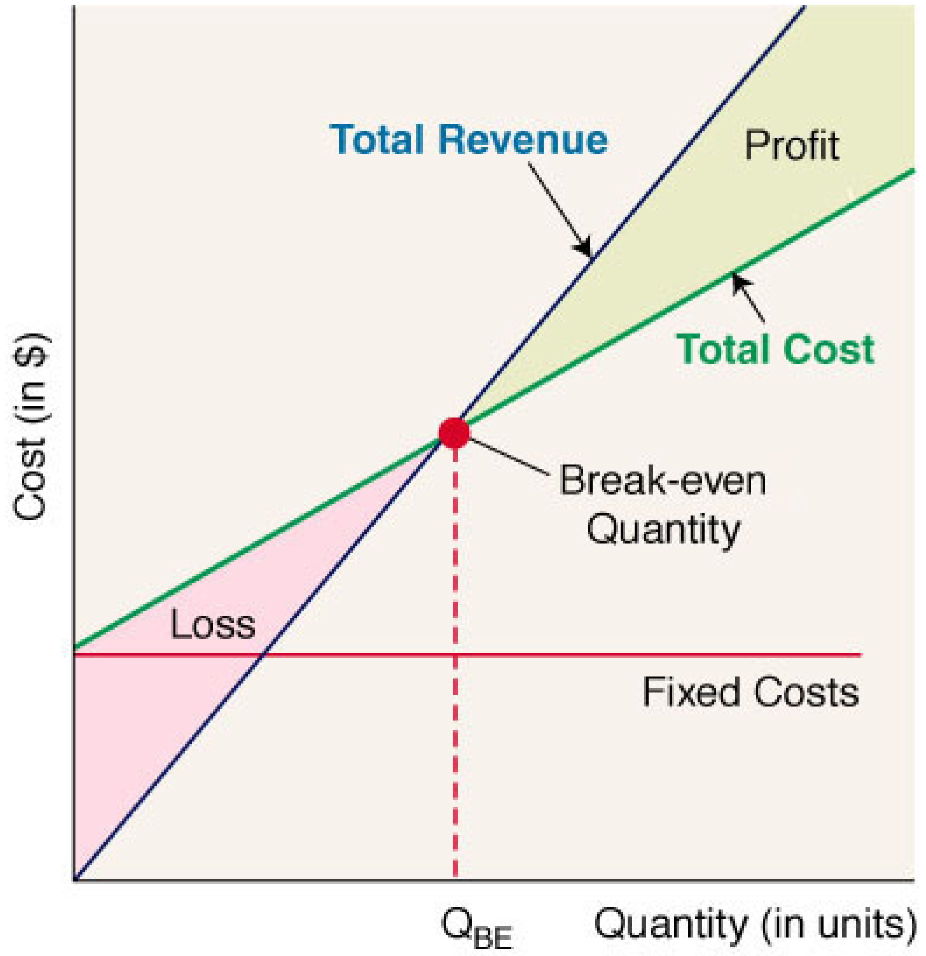
Idea Development

- Existing & target customers
  - Customer surveys & focus groups
- Benchmarking
  - Studying "best in class" companies from your industry or others and comparing their practices & performance to your own
- Reverse engineering
  - Disassembling a competitor's product & analyzing its design characteristics & how it was made
- Suppliers, employees and technical advances

Product Screening

- Operations:
  - Are production requirements consistent with existing capacity?
  - Are the necessary labor skills & raw materials available?
- Marketing:
  - How large is the market niche?
  - What is the long-term potential for the product?
- Finance:
  - What is the expected return on investment?

Break-Even Analysis



Break-Even Analysis

- Total cost = fixed costs + variable costs (quantity):

$$TC = F + (VC)Q$$

- Revenue = selling price (quantity)

$$R = (SP)Q$$

- Break-even point is where total costs = revenue:

$$TC = R \quad or \quad F + (VC)Q = (SP)Q$$

$$or \quad Q = \frac{F}{SP - VC}$$

Break-Even Analysis Example

- A firm estimates that the fixed cost of producing a line of footwear is $52,000 with a$ 9 variable cost for each pair produced.
  - If each pair sells for $25, how many pairs must they sell to break-even?
  - If they sell 4000 pairs at $25 each, how much money will they make?

Example Solved

- Break-even point:

$$Q = \frac{F}{SP - VC} = \frac{52{,}000}{25 -} 50\ pairs$$

$$9$$

- Profit = total revenue – total costs

$$\begin{aligned} P &= (SP)Q - (F + (VC)Q) \\ &= (25)4000 - (52{,}000 + (9)4000) \\ &= 12{,}000 \end{aligned}$$

Preliminary Design & Testing

- General performance characteristics are translated into technical specifications
- Prototypes are built & tested (maybe offered for sale on a small scale)
- Bugs are worked out & designs are refined

Final Design

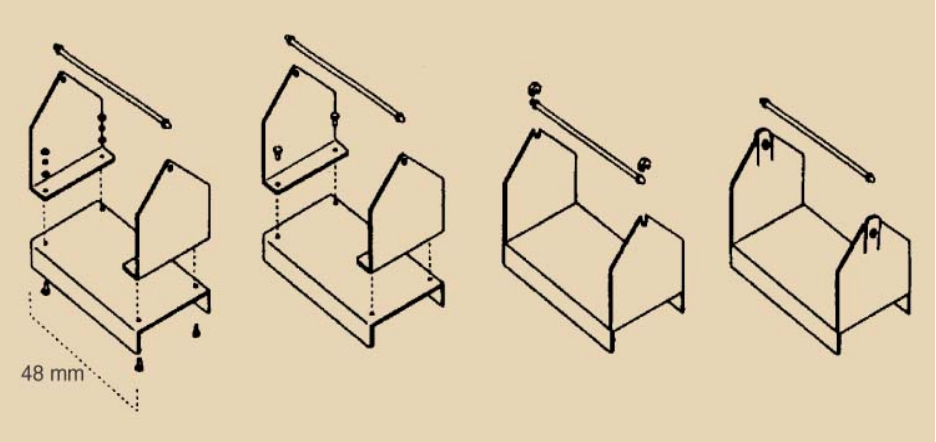
- Specifications are set & then used to:
  - Develop processing and service delivery instructions
  - Guide equipment selection
  - Outline jobs to be performed
  - Negotiate contracts with suppliers and distributors

Design factors

- Product Life Cycle
- Design for Manufacture
- Concurrent Engineering

Design for Manufacture (DMF)

- **Minimize parts**
- **Design parts for multiply applications**
- **Use modular design**
- **Avoid tools**
- **Simplify operations**



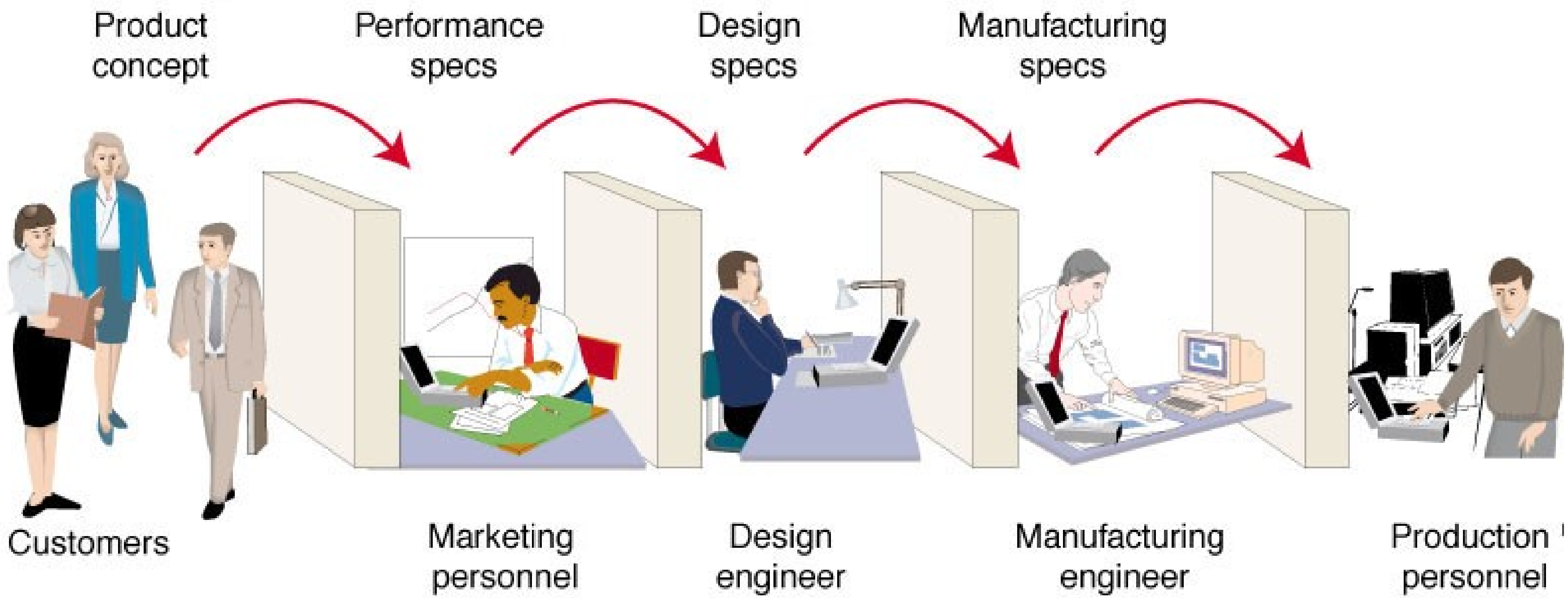
DFM Benefits

- Lower costs:
  - Lower inventories (fewer, standardized components)
  - Less labor required (simpler flows, easier tasks)
- Higher quality:
  - Simple, easy-to-make products means fewer opportunities to make mistakes

Concurrent Engineering

- A design approach that uses multifunctional teams to simultaneously design the product & process
- Replaces a traditional 'over-the-wall' approach where one group does their part & then hands off the design to the next group

Sequential Design



Concurrent Engineering

Concurrent Engineering Benefits

- Representatives from the different groups can better consider trade-offs in cost & design choices as each decision is being made
- Development time is reduced due to less rework (traditionally, groups would argue with earlier decisions & try to get them changed)
- Emphasis is on problem-solving (not placing blame on the ‘other group’ for mistakes)

Process Selection

- **Process selection is based on five considerations**
  - Type of process; range from intermittent to continuous
  - Degree of vertical integration
  - Flexibility of resources
  - Mix between capital & human resources
  - Degree of customer contact

- **Process types can be:**
  - Project Process
  - Batch Process
  - Line Process
  - Continuous Process